Cost Optimization Solutions For E-Commerce Vendors

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Introduction

- Since the introduction of e-commerce, the retail industry has sky rocketed.
- Although this is advantageous for the customer, the vendor also has to look at his/her own costs.
- To win over the customer, the delivery process must be sped up by using methods such as Same Day Delivery (SDD), all the while ensuring the vendor doesn’t go over-budget.
- An effective way to satisfy the customer as well as balancing the costs is by using customized logistic optimization methods.
- One such method is the popular Differential Evolution Algorithm.
- The commodity chosen for this project is milk, which being a a perishable product, makes time a major constraint.

Review Of Literature

How Retail can Scale Fast Using Logistics Optimization

- With many vendors opting to sell online, the customers have an even wider range to choose from.
- Winning over customers becomes essential, by speeding up delivery all the while balancing the accounts.
- This requires logistics optimizations, by using methods such as:

1. Optimization Technology – This involves mapping out the entire supply chain and collecting information such as distribution costs and the size of the orders.
2. Fulfilling Same Day Delivery – This means that the order should be ready to ship within 3-4 hours of placing the order. Combined with Last Mile Delivery, this speeds up the entire process.

Review Of Literature

3. Developing Cost Effective SCM – By scanning and automating as many steps as possible, results in massive savings.
4. Managing live delivery with end-to-end visibility – Optimization involves using AI and ML to select the best path for delivery along with choosing the nearest driver.

Review Of Literature

- **Supply Chain Management in a Dairy Industry – A Case Study**
- A supply chain is a model where various business entities come together to address the issue of material and information flow.
- By using a reference model, an attempt has been made to develop a co-ordinated supply chain model with procurement, production and distribution systems.
- A multi-objective function is then formulated to minimize cost subject to supplies, plant and distribution capacities, production and distribution through put limits and customs demand requirements.
- Total cost includes fixed costs of production and distribution, variable costs of production, distribution and transportation. This model consists of four echelons namely Suppliers, Plants, Distribution Centres (DCs), and Customer zones (CZs).

System Model

Mathematical Model

- Material Cost = $\Sigma A_{v,m} \times C_{v,m} \ \forall v,m$; where v = vendor, m = material
- Production Cost = $\Sigma A_{g,f} \times C_{g,f} \ \forall f,g$; where g = goods, f = facility
- Transportation Cost = $\Sigma A_{m,v,f} \times C_{m,v,f} \ \forall v,f,m + \Sigma A_{g,f,w} \times C_{g,w,f} \ \forall f,w,g + \Sigma A_{g,w,c} \times C_{g,w,c} \ \forall w,c,g$ where v = vendor, f = facility, m = material, w = warehouse, g = goods, c = customer
- Objective Function = Min $(\Sigma A_{v,m} \times C_{v,m} \ \forall v,m + \Sigma A_{g,f} \times C_{g,f} \ \forall f,g + \Sigma A_{m,v,f} \times C_{m,v,f} \ \forall v,f,m + \Sigma A_{g,f,w} \times C_{g,w,f} \ \forall f,w,g + \Sigma A_{g,w,c} \times C_{g,w,c} \ \forall w,c,g + IC)$

System Model

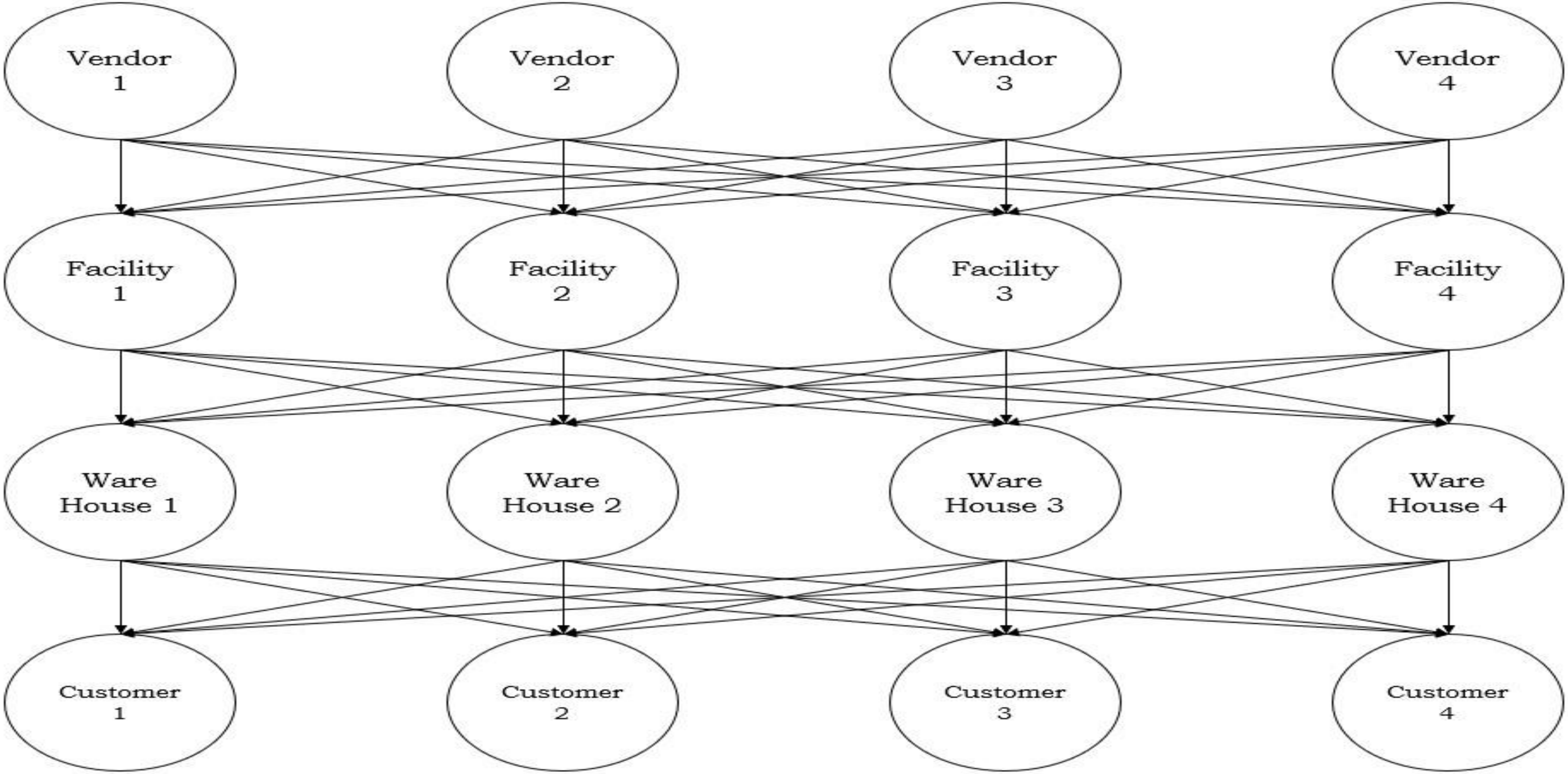
Mathematical Model

- Limits

1. $0 \leq A_{m,v,f} \leq A_{m,v,f}(max)$
2. $0 \leq A_{g,f,w} \leq A_{g,f,w}(max)$
3. $0 \leq A_{g,w,c} \leq A_{g,w,c}(max)$
4. $A_{g,f}(min) \leq A_{g,f} \leq A_{g,f}(max)$

- Restrictions

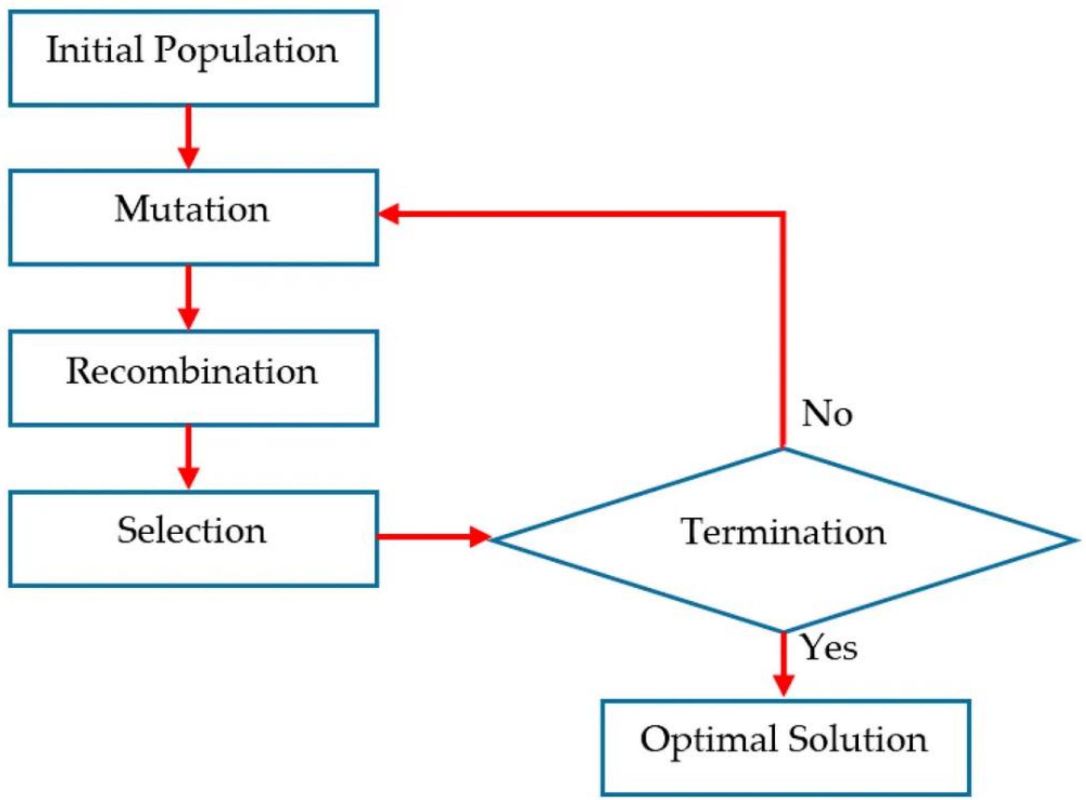
1. $A_{m,v,f} = (\text{Amount sold})_{v,f}\ \forall v,m,f$
2. $A_{g,w,c} = (\text{Amount sold})_{g,c}\ \forall w,c,g$



Network Model

Differential Evolution

- Population based meta heuristic search algorithm.
- Optimization done on an iterative basis.
- DE does not make any assumptions about the underlying data.
- Can explore the entire data very speedily.



Differential Evolution Working

Step 1: Initialize the algorithm to G = 0, where G is the total number of iterations.

Step 2: Generate 3 random individuals from the initial population, such that they are distinct and not equal to one another.

Step 3: Next, form the donor vector using these individuals using the mutation function.

Step 4: Perform crossover. This is done by developing a trial vector which is formed such that, if the index of the donor vector is less than or equal to the crossover probability then choose the donor vector. Else choose the original individual.

Step 5: Evaluate if the objective value of the trial vector is less than or equal to the objective value of the individual. If so then replace the individual with the trial vector, else keep it.

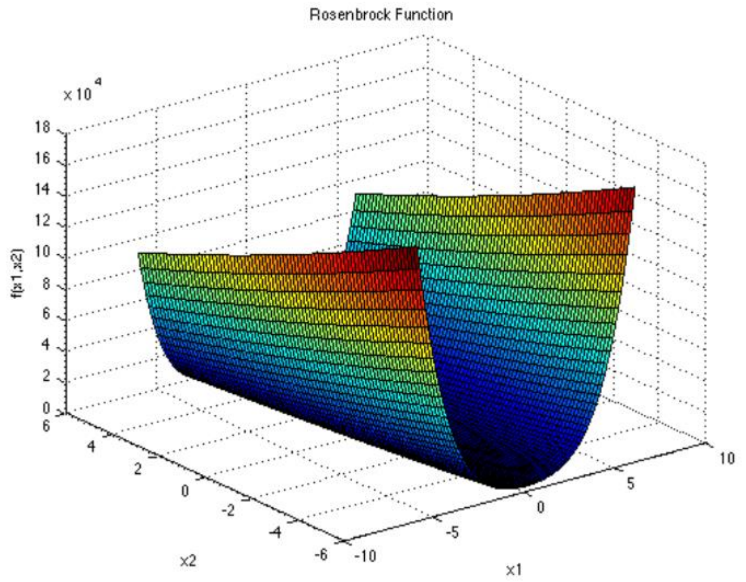
Step 6: Check if the stopping criterion has been met. If met then stop, else go to step 2.

System Implementation

- **Using the Rosenbrock Function**

The Rosenbrock Function is a unimodal function which is popularly used for testing optimization algorithms such as DE. It is also known as the Banana Function. It is given as:

$$\mathbf{F(x) = \sum_{i=1}^{d-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]}$$



System Implementation using the Rosenbrock Function

```
Iteration: 7 f([[0.97841 0.96458]]) = 0.00580
Iteration: 12 f([[1.02934 1.05441]]) = 0.00350
Iteration: 14 f([[1.02959 1.05721]]) = 0.00168
Iteration: 27 f([[0.95945 0.92093]]) = 0.00166
Iteration: 29 f([[0.96362 0.92824]]) = 0.00133
Iteration: 33 f([[0.96453 0.93014]]) = 0.00126
Iteration: 34 f([[1.02237 1.04274]]) = 0.00112
Iteration: 35 f([[1.02259 1.04614]]) = 0.00053
Iteration: 36 f([[0.99409 0.98853]]) = 0.00004
Iteration: 45 f([[1.00314 1.00583]]) = 0.00003
Iteration: 47 f([[0.99619 0.99218]]) = 0.00002
Iteration: 48 f([[1.00028 1.00065]]) = 0.00000

Solution: f([[1.00028 1.00065]]) = 0.00000
```

System Implementation

Excluding Inventory Cost

```
Solution: f([[-1.45067  0.24073]]) = 0.03000

Process finished with exit code 0
```

Including Inventory Cost

```
Solution: f([[ 1.60935 -3.22577]]) = 0.03000

Process finished with exit code 0
```

Conclusion

- For this project, I have studied the Supply Chain Management and the effect of e-commerce on the retail sector. Along with this, a basic cost optimization model has been developed. The objective function has been developed, which is the minimization of the total cost. This minimization is to be achieved using an iterative algorithm.
- I have used the very widely popular Differential Evolution for this. I have implemented the algorithm using the Rosenbrock Function first, before implementing it using the collected real world data.
- In conclusion, such a solution strategy is a very viable option for such optimization problems.